

The Foundations: Logic and Proofs

Chapter 1, Part III: Proofs

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Summary

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Rules of Inference

Section 1.6

Section Summary₁

- Valid Arguments in Propositional Logic
- Rules of Inference for Propositional Logic
- Using Rules of Inference to Build Arguments
- Resolution
- Fallacies
- Rules of Inference for Quantified Statements
- Combining Rules of Inference for Propositions and Quantified Statements

Revisiting the Socrates Example

We have the two **premises**:

- “All men are mortal.”
- “Socrates is a man.”

And the **conclusion**:

- “Socrates is mortal.”

How do we get the conclusion from the premises?

1.6.2 Valid Arguments in Propositional Logic

We can express the premises (above the line) and the conclusion (below the line) in predicate logic as an **argument**:

$$\frac{\forall x (Man(x) \rightarrow Mortal(x)) \quad Man(Socrates)}{\therefore Mortal(Socrates)}$$

where $\therefore$ is the symbol that denotes “therefore.”

This is a **valid argument**.

1.6.2 Valid Arguments in Propositional Logic

Definition 1: An **argument** in propositional logic is a sequence of propositions. All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**. An argument is **valid** if the truth of all its premises implies that the conclusion is true.

An **argument form** in propositional logic is a sequence of compound propositions involving propositional variables. An **argument form** is **valid** if no matter which propositions are substituted for the propositional variables in its premises, if the premises are all true, then the conclusion is true.

1.6.2 Valid Arguments in Propositional Logic

In other words, an **valid argument form** with premises $p_1, p_2, \dots, p_n$ and conclusion q is valid

if and only if $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a **tautology**.

The **key to showing that** an argument in propositional logic is valid is to show that its argument form is valid.

We will now **develop methods** for accomplishing this task (to show that argument forms are valid).

1.6.3 Rules of Inference for Propositional Logic

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Modus Ponens (MP)

$$p \rightarrow q$$

$$\frac{p}{\therefore q}$$

Corresponding Tautology:

$$(p \wedge (p \rightarrow q)) \rightarrow q$$

Example:

Let p be “It is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“It is snowing.”

“Therefore , I will study discrete math.”

Modus Ponens (MP)

If $\sqrt{2} > \frac{3}{2}$, then $(\sqrt{2})^2 > \left(\frac{3}{2}\right)^2$. We know that $\sqrt{2} > \frac{3}{2}$.

Therefore, $(\sqrt{2})^2 = 2 > \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

Is the argument valid?

Does the conclusion must be true?

The argument **is valid**: modus ponens inference rule.

We **cannot conclude** that the conclusion is true, since one of its premises, $\sqrt{2} > \frac{3}{2}$ is false.

Indeed, in this case **the conclusion is false**, since $2 \not> \left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2.25$.

Modus Tollens (MT)

$$p \rightarrow q$$

$$\frac{\neg q}{\therefore \neg p}$$

Corresponding Tautology:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

Example:

Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore , it is not snowing.”

Hypothetical Syllogism (HS)

$$p \rightarrow q$$

$$q \rightarrow r$$

$$\hline \therefore p \rightarrow r$$

Corresponding Tautology:

$$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

Example:

Let p be “it snows.”

Let q be “I will study discrete math.”

Let r be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore , If it snows, I will get an A.”

Disjunctive Syllogism (DS)

$$p \vee q$$

$$\frac{\neg p}{\therefore q}$$

Corresponding Tautology:

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English.”

“I will study discrete math or I will study English.”

“I will not study discrete math.”

“Therefore , I will study English.”

Addition

$$\frac{p}{\therefore p \vee q}$$

Corresponding Tautology:

$$p \rightarrow (p \vee q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math or I will visit Las Vegas.”

Simplification

$$\frac{p \wedge q}{\therefore p}$$

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English.”

“I will study discrete math and English”

“Therefore, I will study discrete math.”

Conjunction

p

q

$\therefore p \wedge q$

Corresponding Tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English.”

“I will study discrete math.”

“I will study English.”

“Therefore , I will study discrete math and I will study English.”

Resolution

$$\neg p \vee r$$

$$\frac{p \vee q}{\therefore q \vee r}$$

$$\therefore q \vee r$$

Corresponding Tautology:

$$((\neg p \vee r) \wedge (p \vee q)) \rightarrow (q \vee r)$$

Example:

Let p be “I will study discrete math.”

Let r be “I will study English.”

Let q be “I will study databases.”

“I will not study discrete math or I will study English.”

“I will study discrete math or I will study databases.”

“Therefore, I will study databases or I will study English.”

Resolution plays an important role in AI and is used in Prolog.

Rules of Inference for Propositional Logic

Example: Which rule of inference is used in each argument below?

- Alice is a Math major. Therefore, Alice is either a Math major or a CSI major.
- Jerry is a Math major and a CSI major. Therefore, Jerry is a Math major.
- If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
- If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
- I go swimming or eat an ice cream. I did not go swimming. Therefore, I eat an ice cream.

<i>Rule of Inference</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	Resolution

1.6.4 Using Rules of Inference to Build Valid Arguments

A *valid argument* is a sequence of statements. Each statement is either a *premise* or follows from previous statements by rules of inference. The last statement is called *conclusion*.

A valid argument takes the following form:

$$\begin{array}{c} S_1 \\ S_2 \\ \cdot \\ \cdot \\ \cdot \\ S_n \\ \therefore C \end{array}$$

Using Rules of Inference to Build Valid Arguments₂

When there are many premises, **several rules** of inference are often needed to **show that an argument is valid**, where the steps of arguments are displayed on **separate lines**, with the **reason for each step** explicitly stated.

Using Rules of Inference to Build Valid Arguments₂

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Using Rules of Inference to Build Valid Arguments₃

Example 2:

“It is not sunny this afternoon and it is colder than yesterday.”

“We will go swimming only if it is sunny.”

“If we do not go swimming, then we will take a canoe trip.”

“If we take a canoe trip, then we will be home by sunset.”

Conclusion: “We will be home by sunset.”

Solution:

1. Choose propositional variables:

p : “It is sunny this afternoon.” q : “It is colder than yesterday.”

r : “We will go swimming.” s : “We will take a canoe trip.”

t : “We will be home by sunset.”

2. Translation into propositional logic:

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

3. Construct the Valid Argument:

Using Rules of Inference to Build Valid Arguments₄

Hypotheses: $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s, s \rightarrow t$

Conclusion: t

3. Construct the Valid Argument

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

1.6.5 Resolution

Recall that the rule of inference called **resolution** is based on the tautology:

$$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$$

If we express the hypotheses and the conclusion as **clauses** (possible by CNF, a conjunction of clauses), we can use **resolution** as the only inference rule to build proofs!

A compound proposition is in *conjunctive normal form (CNF)* if it is a conjunction of disjunctions of simple terms.

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee r \vee s) \wedge (\neg p \vee \neg q)$$

1.6.5 Resolution

Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

We can rewrite the premises $(p \wedge q) \vee r$ as two **clauses** (a disjunction of variables or negations of these variables), $p \vee r$ and $q \vee r$.

We can also replace $r \rightarrow s$ by the equivalent clause $\neg r \vee s$.
Using the two clauses $p \vee r$ and $\neg r \vee s$,
we can use resolution to conclude $p \vee s$.

1.6.6 Fallacies

Fallacy = misconception resulting from incorrect argument.

- **Fallacy of affirming the conclusion**

- **Fallacy of denying the hypothesis**

1.6.7 Rules of Inference for Quantified Statements

TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall xP(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall xP(x)}$	Universal generalization
$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists xP(x)}$	Existential generalization

Rules of Inference for Quantified Statements: Universal Instantiation

$$\frac{\forall xP(x)}{\therefore P(c)}$$

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

Therefore, Fido is cuddly.

Rules of Inference for Quantified Statements: Universal Generalization

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

The element c that we select must be an **arbitrary**, and not a specific, element of the domain.

Universal generalization is used implicitly in many proofs in mathematics and is seldom mentioned explicitly.

However, the **error** of adding unwarranted assumptions about the arbitrary element c when universal generalization is used is all too common in incorrect reasoning.

Rules of Inference for Quantified Statements: Existential Instantiation

$$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$$

Example:

“There is someone who got an A in the course.”

“Let’s call her a and say that a got an A”

Rules of Inference for Quantified Statements: Existential Generalization

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Example:

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

Rules of Inference for Quantified Statements

Example 1: Using the rules of inference, construct a valid argument to show that

“John Smith has two legs”

is a consequence of the premises:

“Every man has two legs.” “John Smith is a man.”

Solution:

Let $M(x)$ denote “ x is a man” and $L(x)$ “ x has two legs” and let John Smith be a member of the domain.

Valid Argument:

Step	Reason
1. $\forall x(M(x) \rightarrow L(x))$	Premise
2. $M(J) \rightarrow L(J)$	UI from (1)
3. $M(J)$	Premise
4. $L(J)$	Modus Ponens using (2) and (3)

Rules of Inference for Quantified Statements

Example 2: Show that the **premises** “A student in this class has not read the book,” and “Everyone in this class passed the first exam” **imply** the **conclusion** “Someone who passed the first exam has not read the book.”

Solution:

- $C(x)$ denote “ x is in this class,”
- $B(x)$ denote “ x has read the book,” and
- $P(x)$ denote “ x passed the first exam.”

Then we translate the premises and conclusion into symbolic form.

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Rules of Inference for Quantified Statements

Valid Argument:

Step	Reason
1. $\exists x(C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	EI from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x(C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	UI from (4)
6. $P(a)$	MP from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conj from (6) and (7)
9. $\exists x(P(x) \wedge \neg B(x))$	EG from (8)

Rules of Inference for Quantified Statements: Returning to Socrates

$$\forall x (Man(x) \rightarrow Mortal(x))$$

$$\frac{Man(Socrates)}{\therefore Mortal(Socrates)}$$

Valid Argument

Step	Reason
1. $\forall x (Man(x) \rightarrow Mortal(x))$	Premise
2. $Man(Socrates) \rightarrow Mortal(Socrates)$	UI from (1)
3. $Man(Socrates)$	Premise
4. $Mortal(Socrates)$	MP from (2) and (3)

1.6.8 Combining Rules of Inference for Propositions and Quantified Statements

These inference rules are frequently used and combine propositions and quantified statements:

- Universal Modus Ponens

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore Q(a) \end{array}$$

- Universal Modus Tollens

$$\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ \neg Q(a), \text{ where } a \text{ is a particular element in the domain} \\ \hline \therefore \neg P(a) \end{array}$$

Section Summary₁

- Valid Arguments in Propositional Logic
- Rules of Inference for Propositional Logic
- Using Rules of Inference to Build Arguments
- Resolution
- Fallacies
- Rules of Inference for Quantified Statements
- Combining Rules of Inference for Propositions and Quantified Statements

Homework after class

1.6 Rules of Inference

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Introduction to Proofs

Section 1.7

Section Summary₂

Some Terminology

Understanding How Theorems Are Stated

Methods of Proving Theorems

Direct Proofs

Indirect Proofs

- Proof by Contraposition
- Proof by Contradiction

Mistakes in Proofs

1.7.1 Introduction

In this section we introduce the **notion** of a proof and describe **methods** for constructing proofs.

A **proof** can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems.

Using these ingredients and rules of inference, the **final step** of the proof establishes the truth of the statement being proved.

1.7.1 Introduction

Now, we move from **formal proofs** (e.g. Section 1.6) of theorems toward more **informal proofs**.

In math, CS, and other disciplines, **informal proofs** which are generally shorter, are generally used.

- More than one rule of inference are often used in a step.
- Steps may be skipped.
- The rules of inference used are not explicitly stated.
- Easier for to understand and to explain to people.
- But it is also easier to introduce errors.

Proofs have many practical **applications**:

- verification that computer programs are correct
- establishing that operating systems are secure
- showing that system specifications are consistent

1.7.2 Some Terminology

A *theorem* is a statement that can be shown to be true using:

- definitions
- other theorems
- *axioms* (statements which are given as true)
- rules of inference

Theorems can also be referred to as facts or results.

Less important theorems are sometimes called *propositions*.

A *lemma* is a ‘helping theorem’ or a result which is needed to prove a theorem.

A *corollary* is a result which follows directly from a theorem.

A *conjecture* is a statement that is being proposed to be true. Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false.

1.7.3 Understanding How Theorems Are Stated

Many theorems assert that **a property holds for all elements** in a domain, such as the integers, the real numbers.

Often the universal quantifier (needed for a precise statement of a theorem) is **omitted by** standard mathematical convention.

For example, the statement:

“If $x > y$, where x and y are **positive** real numbers, then $x^2 > y^2$ ”

really means

“For **all positive** real numbers x and y , if $x > y$, then $x^2 > y^2$.”

1.7.4 Method of Proving Theorems

Many theorems have the **form**:

$$\forall x(P(x) \rightarrow Q(x))$$

Our **goal** is to show that $P(c) \rightarrow Q(c)$ is true, where c is an arbitrary element of the domain.

Then, by universal generalization the truth of the original formula follows.

So, we **must prove** something of the form: $p \rightarrow q$

We need **only show that** q is true if p is true.

Method of Proving Theorems: Proving Conditional Statements: $p \rightarrow q$

When you read proofs, you will often find the words “obviously” or “clearly.”

Vacuous Proof: If we know p is false then $p \rightarrow q$ is true as well.

“If I am both rich and poor then $2 + 2 = 5$.”

Trivial Proof: If we know q is true, we can prove $p \rightarrow q$ is true as well.

“If it is raining then $1=1$.”

[Even though these *examples seem silly*, both trivial and vacuous proofs are often used in *mathematical induction*, as we will see in Chapter 5)]

1.7.5 Direct Proofs

A direct proof of a conditional statement

$p \rightarrow q$:

Step 1: Assume that p is true.

Step 2: Use rules of inference, axioms, and logical equivalences.

Step 3: Showing that q must also be true.

1.7.5 Direct Proofs

Definition: The integer n is **even** if there exists an integer k such that $n = 2k$, and n is **odd** if there exists an integer k , such that $n = 2k + 1$. **Note that** every integer is either even or odd and no integer is both even and odd.

We will need this basic fact about the integers in some of the example proofs to follow. We will learn more about the integers in Chapter 4.

Direct Proofs: Proving Conditional Statements: $p \rightarrow q$

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Solution: Assume that n is odd. Then $n = 2k + 1$ for an integer k . Squaring both sides of the equation, we get:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1,$$

where $r = 2k^2 + 2k$, an integer.

We have proved that if n is an odd integer, then n^2 is an odd integer.

Proofs

Direct Proofs

Indirect Proofs

- Proof by Contraposition
- Proof by Contradiction

1.7.6 Proof by Contraposition

This method of proof makes use of the equivalence:

$$(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$$

In a proof by contraposition that $p \rightarrow q$, we **assume** $\neg q$ is true, and using axioms, definitions and previously proven theorems, together with inference rules, we **show that** $\neg p$ must be true.

It is a direct proof of the **contrapositive statement**!

From $p \rightarrow q$ we can form new conditional statements .

- $q \rightarrow p$ is the **converse** of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$

When to use each type of proof?

Usually try a direct proof. If it doesn't work, try a proof by contraposition.

Example: Prove that the sum of two rational numbers is rational. (The real number r is *rational* if there exist integers p and q where $q \neq 0$ such that $r = p/q$).

Solution: Assume r and s are two rational numbers. Then there must be integers p, q and also t, u such that

$$r = p / q, \quad s = t / u, \quad u \neq 0, \quad q \neq 0$$
$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu} = \frac{v}{w} \quad \begin{array}{l} \text{where } v = pu + qt \\ w = qu \neq 0 \end{array}$$

Thus the sum is rational. (The **direct proof** succeeded!)

When to use each type of proof?

Prove that for any integer number n , if n^2 is odd, then n is odd.

Trying a direct proof...

Let n be an integer number. Assume that n^2 is odd. We get next that there exists an integer k such that $n^2 = 2k + 1$. Solving for n produces the equation $n = \pm\sqrt{2k + 1}$, which is not very useful to show that n is odd.

Try a prove by contraposition...

Let n be an integer number. Assume n is not odd. This means that n is even, and so there exists an integer k such that $n = 2k$. Thus, $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. So, taking $k' = 2k^2$, we see that $n^2 = 2k'$ and so n^2 is even. This concludes our proof by contraposition.

When to use each type of proof?

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Trying a direct proof...

We first assume that $3n + 2$ is an odd integer. We know that $3n + 2 = 2k + 1$ for some integer k . Thus, $3n + 1 = 2k$, but there does not seem to be any direct way to conclude that n is odd. So, our attempt at a direct proof failed.

Try a prove by contraposition...

Assume n is even. So, $n = 2k$ for some integer k . Thus $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j$ for $j = 3k + 1$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even), then n is odd (not even).

1.7.7 Proof by Contradiction

In a proof by contradiction, we prove that a **proposition** p is true, by showing that there exists a **contradiction** q such that $:\neg p \rightarrow q$.

We can prove that p is true by showing for instance that $:\neg p \rightarrow (r \wedge \neg r)$, for some proposition r .

Example: Prove that $\sqrt{2}$ is irrational.

A direct proof is difficult, as it means to show that there exists no two integer a and b with $b \neq 0$ such that $\sqrt{2} = a/b$. Nonexistence proofs are usually hard.

Let's try a proof by contradiction...

Proof by Contradiction₁

Example: Use a proof by contradiction to give a proof that $\sqrt{2}$ is irrational.

Solution: Suppose $\sqrt{2}$ is rational. Then there exists integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors (see Chapter 4). Then

$$2 = \frac{a^2}{b^2} \qquad 2b^2 = a^2$$

Therefore a^2 must be even. If a^2 is even then a must be even (an exercise). Since a is even, $a = 2c$ for some integer c . Thus,

$$2b^2 = 4c^2 \qquad b^2 = 2c^2$$

Therefore b^2 is even. Again then b must be even as well.

But then 2 must divide both a and b . This contradicts our assumption that a and b have no common factors. We have proved by contradiction that our initial assumption must be false and therefore $\sqrt{2}$ is irrational.

Proof by Contradiction₂

Example: Prove that if you pick 22 days from the calendar, at least 4 must fall on the same day of the week.

Solution: Let p be the proposition “At least 4 of 22 chosen days fall on the same day of the week.” Suppose that $\neg p$ is true. $\neg p$: “no more than 3 of the 22 days fall on the same day of the week.” (r).

Because there are 7 days of the week, we could only have picked 21 days, not 22 days ($\neg r$). That is, we have shown that $\neg p \rightarrow (r \wedge \neg r)$. Consequently, the p is true.

Proof by Contradiction: Biconditional Statements

To prove a theorem that is a **biconditional statement**, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true.

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

Example: Prove the theorem: “If n is an integer, then n is odd if and only if n^2 is odd.”

Solution: We have already shown (previous slides) that both $p \rightarrow q$ and $q \rightarrow p$. Therefore we can conclude $p \leftrightarrow q$.

Sometimes *iff* is used as an **abbreviation** for “if and only if,” as in

“If n is an integer, then n is odd iff n^2 is odd.”

1.7.8 Mistakes in Proofs

What's wrong with the step of “Proof” that $1 = 2$?

Step	Reason
1. $a = b$	Premise
2. $a^2 = a \times b$	Multiply both sides of (1) by a
3. $a^2 - b^2 = a \times b - b^2$	Subtract b^2 from both sides of (2)
4. $(a - b)(a + b) = b(a - b)$	Algebra on (3)
5. $a + b = b$	Divide both sides by $a - b$
6. $2b = b$	Replace a by b in (5) because $a = b$
7. $2 = 1$	Divide both sides of (6) by b

Solution: Step 5. $a - b = 0$ by the premise and division by 0 is undefined.

1.7.9 Just a Beginning

If **direct methods** of proof do not work:

- We may need a clever use of a proof by **contraposition**.
- Or a proof by **contradiction**.
- In the next section, we will see **strategies** that can be used when straightforward approaches do not work.
- In Chapter 5, we will see **mathematical induction** and related techniques.
- In Chapter 6, we will see **combinatorial proofs**

Section Summary₂

Some Terminology

Understanding How **Theorems** Are Stated

Methods of Proving Theorems

Direct Proofs

Indirect Proofs

- Proof by **Contraposition**
- Proof by **Contradiction**

Mistakes in Proofs

Extra exercises

Show that the statement “Every integer is less than its cube” is false by finding a counterexample.

$-2 > (-2)^3 = -8$, so that -2 is not less than its cube, -8 .

Because we have found a counterexample, we know that the statement “Every integer is less than its cube” is false.

(Note that every negative integer is a counterexample to this statement.)

Extra exercises

Using the definitions of even integer and odd integer, give a direct proof that this statement is true for all integers n : If n is odd, then $5n + 3$ is even.

Solution:

Suppose n is odd.

Therefore $n = 2k + 1$ for some integer k

Therefore $5n + 3 = 5(2k + 1) + 3$

Therefore $5n + 3 = 10k + 8$

Therefore $5n + 3 = 2(5k + 4)$

Therefore $5n + 3$ is even

given

definition of odd integer

substitution of $2k + 1$ for n

algebra

algebra

$5n + 3$ is an integer multiple of 2

We give a proof using two columns, one for statements and one for reasons.

Extra exercises

Suppose we need to prove that this statement is true for all integers n , using the definitions of even integer and odd integer: **If $7n-5$ is odd, then n is even.**

direct proof

Suppose $7n - 5$ is odd.

Therefore $7n - 5 = 2k + 1$ for some integer k .

Therefore $7n = 2k + 6 = 2(k + 3)$.

Therefore $n = \frac{2(k+3)}{7}$.

proof by

contraposition

Suppose n is not even. Therefore n is odd.

Therefore $n = 2k + 1$ for some integer k .

Therefore $7n - 5 = 7(2k + 1) - 5 = 14k + 2 = 2(7k + 1)$.

Therefore $7n - 5$ is even.

Extra exercises

Suppose a , b , and c are odd integers. Prove that the roots of $ax^2 + bx + c = 0$ are not rational.

Solution:

In general, it is not very easy to give a direct proof that numbers are not rational. Here we would need to show that neither root can be written as a quotient of integers. It might be easier to try a proof by contradiction because we can make the assumption that the roots *are* rational and hence can be written as fractions. That is, there is a very specific form in which the roots can be written.

Suppose the roots of $ax^2 + bx + c = 0$ are rational — say the roots are e/f and g/h . Then we can write $\left(x - \frac{e}{f}\right)\left(x - \frac{g}{h}\right) = 0$. Equivalently, this is $\left(\frac{fx - e}{f}\right) \cdot \left(\frac{hx - g}{h}\right) = 0$, or $(fx - e)(hx - g) = 0$.

Multiplying together the two terms yields:

$$(fh)x^2 - (fg + eh)x + (eg) = 0.$$

Because this must have the form $ax^2 + bx + c = 0$, the corresponding coefficients must be equal:

$$fh = a, \quad -(fg + eh) = b, \quad eg = c.$$

Therefore f and h must be odd because their product is the odd integer a . Likewise, e and g must be odd because their product is the odd integer c . But this forces $fg + eh$ to be even because it is the sum of the odd integers fg and eh . Therefore $b = -(fg + eh)$ must be even, a contradiction of the fact that b is odd.

Extra exercises

Suppose you are examining a conjecture of the form $\forall x(P(x) \rightarrow Q(x))$. If you are looking for a counterexample,

you need to find a value x such that:

- A) $P(x)$ and $Q(x)$ are true.
- B) $P(x)$ and $Q(x)$ are false.
- C) $Q(x)$ is true and $P(x)$ is false.
- D) $P(x)$ is true and $Q(x)$ is false.

D) **[CORRECT]** In order to have a statement of the form $\forall x(P(x) \rightarrow Q(x))$ false, you need a value x such that $P(x) \rightarrow Q(x)$ is false. To have an implication be false, you must have the hypothesis true and the conclusion false. That is, you need to find a value x such that $P(x)$ is true and $Q(x)$ is false.

Homework after class

1.7 Introduction to Proofs

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Proof Methods and Strategy

Section 1.8

Section Summary₃

Proof by Cases

Existence Proofs

- Constructive
- Nonconstructive

Disproof by Counterexample

Nonexistence Proofs

Uniqueness Proofs

Proof Strategies

Proving Universally Quantified Assertions

Open Problems

1.8.2 Exhaustive Proof and Proof by Cases

To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

Use the tautology

$$\begin{aligned} & \left[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q \right] \leftrightarrow \\ & \left[(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q) \right] \end{aligned}$$

Each of the implications $p_i \rightarrow q$ is a *case*.

An exhaustive proof is a special type of proof by cases where each case involves checking a single example.

1.8.2 Exhaustive Proof and Proof by Cases₁

Example: Let $a @ b = \max\{a, b\} = a$ if $a \geq b$, otherwise
 $a @ b = \max\{a, b\} = b$.

Show that for all real numbers a, b, c . $(a @ b) @ c = a @ (b @ c)$

Proof: Let a, b , and c be arbitrary real numbers.

Then one of the following 6 cases must hold.

$$a \geq b \geq c; a \geq c \geq b; b \geq a \geq c; b \geq c \geq a; c \geq a \geq b; c \geq b \geq a$$

Case 1: $a \geq b \geq c$

$$(a @ b) = a, a @ c = a, b @ c = b$$

$$\text{Hence } (a @ b) @ c = a = a @ (b @ c)$$

A complete proof requires that the equality be shown to hold for all 6 cases. But the proofs of the remaining cases are similar.

1.8.2 Exhaustive Proof and Proof by Cases₂

Example: Prove that if n is integer then $n^2 \geq n$.

We split the proof into three cases.

- Case (i) $n = 0$.

In this case, $n^2 = 0^2 = 0 = n$.

- Case (ii) $n \geq 1$.

In this case, when we multiply both sides of $n \geq 1$ by n we obtain $n \cdot n \geq n \cdot 1$. This implies $n^2 \geq n$.

- Case (iii) $n \leq -1$

In this case, $n \leq -1$, but $n^2 \geq 0$. Therefore, $n^2 \geq 0 \geq -1 \geq n$, and so $n^2 \geq n$.

Because the inequality $n^2 \geq n$ holds in all three cases, we can conclude that if n is an integer, then $n^2 \geq n$.

1.8.2 Exhaustive Proof and Proof by Cases₃

Example: Show that if x and y are integers and both $x \cdot y$ and $x + y$ are even, then both x and y are even.

Proof: Use a proof by contraposition. Suppose x and y are not both even. Then, one or both are odd. Without loss of generality, assume that x is odd. Then $x = 2m + 1$ for some integer m .

Case 1: y is even. Then $y = 2n$ for some integer n , so
 $x + y = (2m + 1) + 2n = 2(m + n) + 1$ is odd.

Case 2: y is odd. Then $y = 2n + 1$ for some integer n , so
 $x \cdot y = (2m + 1)(2n + 1) = 2(2m \cdot n + m + n) + 1$ is odd.

We only cover the case where x is odd because the case where y is odd is similar. The use phrase *without loss of generality* indicates this.

1.8.2 Exhaustive Proof and Proof by Cases₄

Common errors with exhaustive proof and proof by cases

- A common error of reasoning is to draw **incorrect conclusions from examples**.
- No matter how many separate examples are considered, a theorem is **not proved by considering examples unless every possible case is covered**.
- No matter how many input values are tested, **unless all input values are tested**, we **cannot conclude that the program always produces the correct output**.

1.8.3 Existence Proofs

Proof of theorems of the form $\exists xP(x)$.

A Constructive Existence Proof:

- Find an explicit value of a , for which $P(a)$ is true.
- Then $\exists xP(x)$ is true by Existential Generalization (EG).

A Nonconstructive Existence Proof:

- We assume no a exists which makes $P(a)$ true and derive a contradiction.

1.8.3 Existence Proofs

Example : A Constructive Existence Proof

Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

Proof:

$$1729 = 10^3 + 9^3 \text{ and } 1729 = 12^3 + 1^3$$

1.8.3 Existence Proofs

Example : A Nonconstructive Existence Proof

Show that there exist irrational numbers x and y such that x^y is rational.

Proof:

We know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$.

If it is rational, we have two irrational numbers x and y with x^y rational, namely $x = \sqrt{2}$ and $y = \sqrt{2}$.

But if $\sqrt{2}^{\sqrt{2}}$ is irrational, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2}\sqrt{2})} = \sqrt{2}^2 = 2$.

1.8.4 Uniqueness Proofs

Some theorems assert the existence of a unique element with a particular property, $\exists!x P(x)$. The two parts of a *uniqueness proof* are

- **Existence**: We show that an element x with the property exists.
- **Uniqueness**: We show that if x and y both have the desired property, then $x = y$.

Example: Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Solution:

- **Existence**: The real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a) + b = -b + b = 0$.
- **Uniqueness**: Suppose that s is a real number such that $as + b = 0$. Then $ar + b = as + b$, where $r = -b/a$. Subtracting b from both sides and dividing by a shows that $r = s$.

1.8.5 Proof Strategies

Choose a method.

1. First try a direct method of proof.
2. If this does not work, try an indirect method (e.g., try to prove the contrapositive).

For whichever method you are trying, choose a strategy.

1. First try *forward reasoning*. Start with the axioms and known theorems and construct a sequence of steps that end in the conclusion. Start with p and prove q , or start with $\neg q$ and prove $\neg p$.
2. If this doesn't work, try *backward reasoning*. When trying to prove q , find a statement p that we can prove with the property $p \rightarrow q$.

Proof Strategies: Backward Reasoning

Given two positive real numbers x and y , their **arithmetic mean** is $(x + y)/2$ and their **geometric mean** is $\sqrt{xy}$. When we compare the arithmetic and geometric means of pairs of distinct positive real numbers, we find that the arithmetic mean is always greater than the geometric mean. [For example, when $x = 4$ and $y = 6$, we have $5 = (4 + 6)/2 > \sqrt{4 \cdot 6} = \sqrt{24}$.] Can we prove that this inequality is always true?

Solution (guide):

$$\begin{aligned}(x + y)/2 &> \sqrt{xy}, \\ (x + y)^2/4 &> xy, \\ (x + y)^2 &> 4xy, \\ x^2 + 2xy + y^2 &> 4xy, \\ x^2 - 2xy + y^2 &> 0, \\ (x - y)^2 &> 0.\end{aligned}$$

(Note that the steps of our backward reasoning will not be part of the final proof. These steps serve as our guide for putting this proof together.)

Proof:

$$\begin{aligned}(x - y)^2 &> 0. \\ x^2 - 2xy + y^2 &> 0, \\ x^2 + 2xy + y^2 &> 4xy, \\ (x + y)^2 &> 4xy, \\ (x + y)^2/4 &> xy, \\ (x + y)/2 &> \sqrt{xy},\end{aligned}$$

Proof Strategies: Backward Reasoning

Example: Suppose that two people play a game taking turns removing, 1, 2, or 3 stones at a time from a pile that begins with 15 stones. The person who removes the last stone **wins the game**. Show that the first player can win the game no matter what the second player does.

Proof: Let n be the last step of the game.

Step n : Player₁ can win if the pile contains **1,2, or 3** stones.

Step $n-1$: Player₂ will have to leave such a pile if the pile that he/she is faced with has **4** stones.

Step $n-2$: Player₁ can leave 4 stones when there are **5,6, or 7** stones left at the beginning of his/her turn.

Step $n-3$: Player₂ must leave such a pile, if there are **8** stones .

Step $n-4$: Player₁ has to have a pile with **9,10, or 11** stones to ensure that there are 8 left.

Step $n-5$: Player₂ needs to be faced with **12** stones to be forced to leave 9,10, or 11.

Step $n-6$: Player₁ can leave 12 stones by removing 3 stones.

Now reasoning forward, the **first player can ensure a win** by **removing 3 stones and leaving 12**.

1.8.6 Looking for Counterexamples

Recall $\exists x \neg P(x) \equiv \neg \forall x P(x)$.

To establish that $\neg \forall x P(x)$ is true (or $\forall x P(x)$ is false), find a c such that $\neg P(c)$ is true or $P(c)$ is false.

In this case c is called a *counterexample* to the assertion $\forall x P(x)$.

Example: “Every positive integer is the sum of the squares of 3 integers.”

- $1 = 0^2 + 0^2 + 1^2$; $2 = 0^2 + 1^2 + 1^2$; $3 = 1^2 + 1^2 + 1^2$;
- $4 = 0^2 + 0^2 + 2^2$; $5 = 0^2 + 1^2 + 2^2$; $6 = 1^2 + 1^2 + 2^2$;
- $7 = 1^2 + 1^2 + 1^2 + 2^2$;

The integer 7 is a counterexample. So the claim is false.

1.8.6 Looking for Counterexamples

Find a counterexample to the statement that the sum of two irrational numbers is also an irrational number.

$$\sqrt{2} + (-\sqrt{2}) = 0$$

To find a counterexample, we will take the sum of $\sqrt{2}$ and $-\sqrt{2}$, which are both irrational numbers. The sum of these numbers is $\sqrt{2} + (-\sqrt{2}) = 0$, which is rational. Note that we could have taken any irrational number and its negative to find a counterexample.

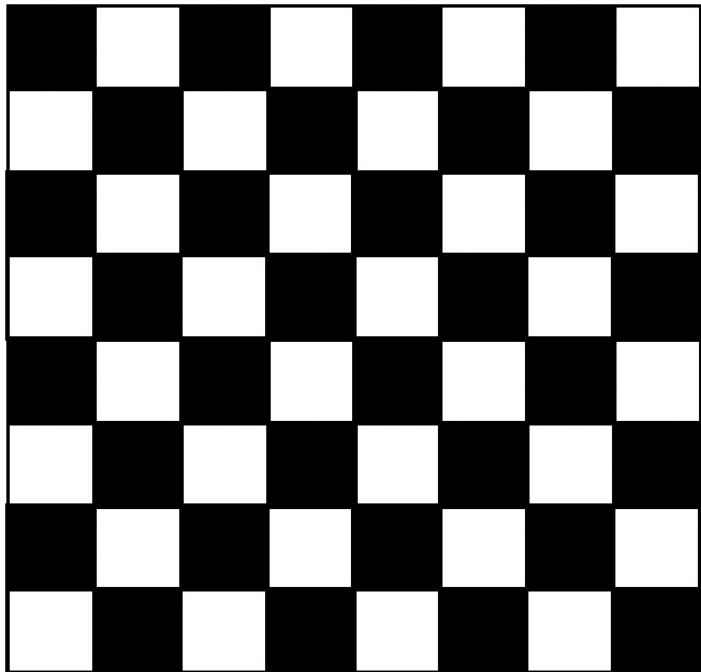
1.8.7 Proof Strategy in Action

- ❑ People formulate **conjectures** on the basis of many types of possible evidence. At other times, conjectures are made based on **intuition** or a **belief** that a result holds.
- ❑ When mathematicians believe **that a conjecture may be true, they try to find a proof.**
 - ❑ If they **cannot find a proof**, they may **look for a counterexample.**
 - ❑ When they cannot find a counterexample, they may switch gears and once **again try to prove** the conjecture.
- ❑ Although many conjectures are quickly settled, a few conjectures resist attack for **hundreds of years** and lead to the development of new parts of mathematics. We will mention a few **famous conjectures** later in this section.

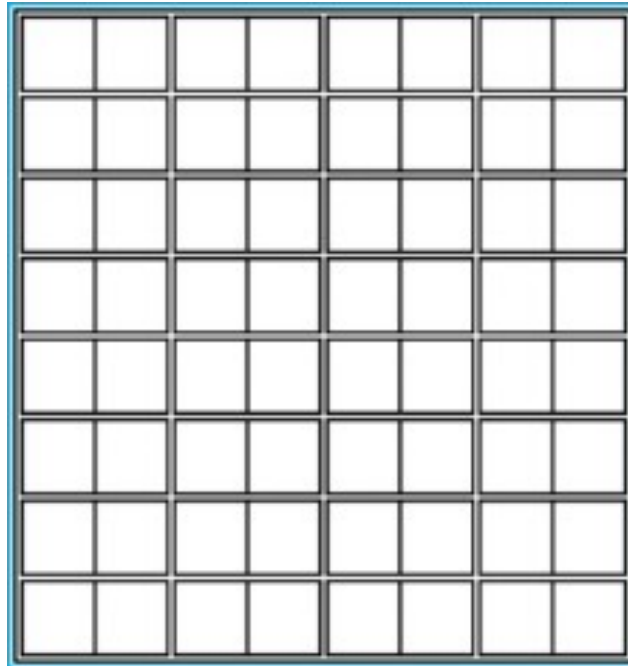
1.8.8 Tilings

Example 1: Can we tile the standard checkerboard using dominos?

Solution: Yes! One example provides a constructive existence proof



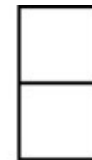
The Standard
Checkerboard



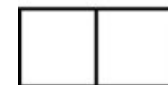
One Possible Solution

- 32 dominoes horizontally or vertically
- some tiles horizontally and some vertically

Two Dominoes

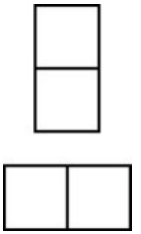


$(2 * 1)$

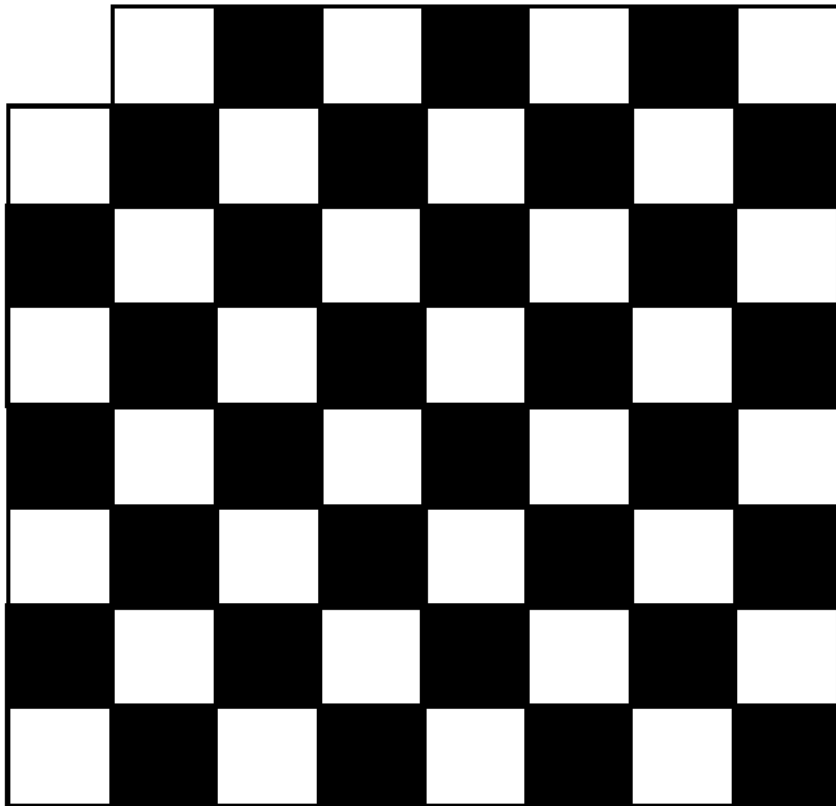


$(1 * 2)$

1.8.8 Tilings₁



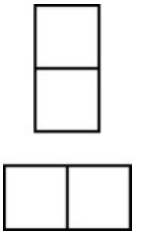
Example 2: Can we tile a checkerboard obtained by removing one of the four corner squares of a standard checkerboard?



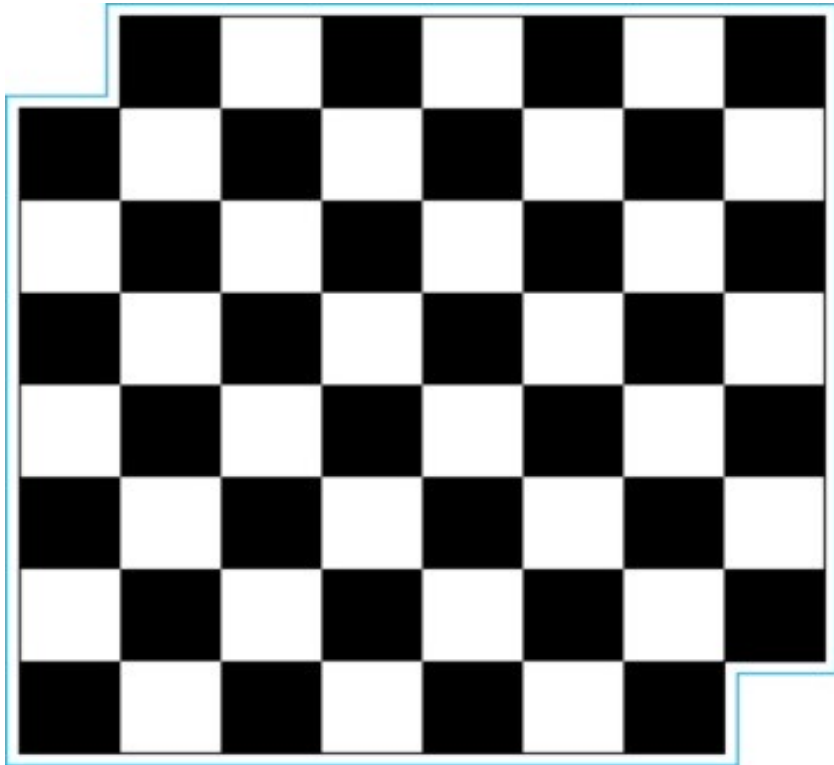
Solution:

- Our checkerboard has $64 - 1 = 63$ squares. Since each domino has two squares, a board with a tiling must have an even number of squares.
- But, the number **63 is not even**.
- We have a **contradiction**.

1.8.8 Tilings₂



Example 3: Can we tile a board obtained by removing both the upper left and the lower right squares of a standard checkerboard?



Nonstandard Checkerboard

Solution:

- There are 62 squares in this board.
- To tile it we need 31 dominos.
- *Key fact:* Each domino covers one black and one white square.
- Therefore the tiling covers 31 black squares and 31 white squares.
- Our board has either 30 black squares and 32 white squares or 32 black squares and 30 white squares.
- **Contradiction!**

1.8.9 The Role of Open Problems

Unsolved problems have motivated much work in mathematics. **Fermat's Last Theorem** was conjectured more than 300 years ago. It has only recently been finally solved.

Fermat's Last Theorem: The equation $x^n + y^n = z^n$ has no solutions in integers x , y , and z , with $xyz \neq 0$ whenever n is an integer with $n > 2$.

A proof was found by Andrew Wiles in the 1990s.

1.8.9 The Role of Open Problems

The $3x + 1$ Conjecture: Let T be the transformation that sends an even integer x to $x/2$ and an odd integer x to $3x + 1$. For all positive integers x , when we repeatedly apply the transformation T , we will eventually reach the integer 1.

For example, starting with $x = 13$:

$$T(13) = 3 \cdot 13 + 1 = 40, T(40) = 40/2 = 20, T(20) = 20/2 = 10,$$

$$T(10) = 10/2 = 5, T(5) = 3 \cdot 5 + 1 = 16, T(16) = 16/2 = 8,$$

$$T(8) = 8/2 = 4, T(4) = 4/2 = 2, T(2) = 2/2 = 1$$

The conjecture has been verified using computers up to $5.48 \cdot 10^{18}$.

1.8.10 Additional Proof Methods

Later we will see many other proof methods:

- **Mathematical induction**, which is a useful method for proving statements of the form $\forall n P(n)$, where the domain consists of all positive integers.
- **Structural induction**, which can be used to prove such results about recursively defined sets.
- **Cantor diagonalization** is used to prove results about the size of infinite sets.
- **Combinatorial proofs** use counting arguments.

Section Summary₃

Proof by Cases

Existence Proofs

- Constructive
- Nonconstructive

Disproof by Counterexample

Nonexistence Proofs

Uniqueness Proofs

Proof Strategies

Proving Universally Quantified Assertions

Open Problems

Extra exercises

Prove that there is only one pair of positive integers that is a solution to $3x^2 + 2y^2 = 30$.

The two terms on the left side of the equation each involve a square, and hence the sum exceeds 30 for small values of x and y . In particular, in order to have a solution, we must have $x \leq 3$ and $y \leq 4$.

The number of such pairs is sufficiently small that we can use an exhaustive proof, considering the cases (1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4).

The only solution among these twelve pairs is (2,3). That is, the only integer solution to $3x^2 + 2y^2 = 30$ is $x = 2$, $y = 3$.

Extra exercises

Prove that the square of every odd integer ends in 1, 5, or 9.

Solution:

Every odd integer n can be written as $n = 10k + r$ where $r = 1, 3, 5, 7, 9$. (For example, $163 = 10 \cdot 16 + 3$ and $7 = 10 \cdot 0 + 7$.) Examine each of these five cases separately, writing each square as “multiple of 10, plus an integer x ”. The integer x will be the units’ digit.

$$n = 10k + 1: n^2 = 100k^2 + 20k + 1 = 10(10k^2 + 2k) + 1, \text{ which ends in } 1.$$

$$n = 10k + 3: n^2 = 100k^2 + 60k + 9 = 10(10k^2 + 6k) + 9, \text{ which ends in } 9.$$

$$n = 10k + 5: n^2 = 100k^2 + 100k + 25 = 10(10k^2 + 10k + 2) + 5, \text{ which ends in } 5.$$

$$n = 10k + 7: n^2 = 100k^2 + 140k + 49 = 10(10k^2 + 14k + 4) + 9, \text{ which ends in } 9.$$

$$n = 10k + 9: n^2 = 100k^2 + 180k + 81 = 10(10k^2 + 18k + 8) + 1, \text{ which ends in } 1.$$

Therefore, in each of the five possible cases the square ends in 1, 5, or 9.

Extra exercises

Suppose a and b are integers such that $2a = b^2 + 3$. Prove that a is the sum of three squares.

Solution:

We first observe that $b^2 + 3$ is even — it is equal to $2a$. Because 3 is odd, b^2 must also be odd. Therefore b must be odd and we can write $b = 2n + 1$. Therefore

$$2a = b^2 + 3 = (2n + 1)^2 + 3 = 4n^2 + 4n + 4.$$

Dividing by 2, we have $a = 2n^2 + 2n + 2$. We need to examine $2n^2 + 2n + 2$ and try to write it as the sum of three squares.

If we write $2n^2$ as $n^2 + n^2$, we have

$$a = n^2 + n^2 + 2n + 2.$$

This gives two squares, but the remaining terms, $2n + 2$, does not appear to be a square. But if we write this as $a = n^2 + n^2 + 2n + 1 + 1$ we obtain a as sum of three squares:

$$a = n^2 + (n^2 + 2n + 1) + 1 = n^2 + (n + 1)^2 + 1.$$

Homework after class

1.8 Proof Methods and Strategy

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Key Terms and Results

TERMS

proposition: a statement that is true or false

propositional variable: a variable that represents a proposition

truth value: true or false

$\neg p$ (negation of p): the proposition with truth value opposite to the truth value of p

logical operators: operators used to combine propositions

compound proposition: a proposition constructed by combining propositions using logical operators

truth table: a table displaying all possible truth values of propositions

$p \vee q$ (disjunction of p and q): the proposition “ p or q ,” which is true if and only if at least one of p and q is true

$p \wedge q$ (conjunction of p and q): the proposition “ p and q ,” which is true if and only if both p and q are true

$p \oplus q$ (exclusive or of p and q): the proposition “ p XOR q ,” which is true when exactly one of p and q is true

$p \rightarrow q$ (p implies q): the proposition “if p , then q ,” which is false if and only if p is true and q is false

converse of $p \rightarrow q$: the conditional statement $q \rightarrow p$

contrapositive of $p \rightarrow q$: the conditional statement $\neg q \rightarrow \neg p$

inverse of $p \rightarrow q$: the conditional statement $\neg p \rightarrow \neg q$

$p \leftrightarrow q$ (biconditional): the proposition “ p if and only if q ,” which is true if and only if p and q have the same truth value

bit: either a 0 or a 1

Boolean variable: a variable that has a value of 0 or 1

bit operation: an operation on a bit or bits

bit string: a list of bits

bitwise operations: operations on bit strings that operate on each bit in one string and the corresponding bit in the other string

logic gate: a logic element that performs a logical operation on one or more bits to produce an output bit

logic circuit: a switching circuit made up of logic gates that produces one or more output bits

tautology: a compound proposition that is always true

contradiction: a compound proposition that is always false

contingency: a compound proposition that is sometimes true and sometimes false

consistent compound propositions: compound propositions for which there is an assignment of truth values to the variables that makes all these propositions true

satisfiable compound proposition: a compound proposition for which there is an assignment of truth values to its variables that makes it true

logically equivalent compound propositions: compound propositions that always have the same truth values

predicate: part of a sentence that attributes a property to the subject

propositional function: a statement containing one or more variables that becomes a proposition when each of its variables is assigned a value or is bound by a quantifier

domain (or universe) of discourse: the values a variable in a propositional function may take

$\exists x P(x)$ (existential quantification of $P(x)$): the proposition that is true if and only if there exists an x in the domain such that $P(x)$ is true

$\forall x P(x)$ (universal quantification of $P(x)$): the proposition that is true if and only if $P(x)$ is true for every x in the domain

logically equivalent expressions: expressions that have the same truth value no matter which propositional functions and domains are used

free variable: a variable not bound in a propositional function

bound variable: a variable that is quantified

scope of a quantifier: portion of a statement where the quantifier binds its variable

argument: a sequence of statements

argument form: a sequence of compound propositions involving propositional variables

premise: a statement, in an argument, or argument form, other than the final one

conclusion: the final statement in an argument or argument form

valid argument form: a sequence of compound propositions involving propositional variables where the truth of all the premises implies the truth of the conclusion

valid argument: an argument with a valid argument form

rule of inference: a valid argument form that can be used in the demonstration that arguments are valid

fallacy: an invalid argument form often used incorrectly as a rule of inference (or sometimes, more generally, an incorrect argument)

circular reasoning or begging the question: reasoning where one or more steps are based on the truth of the statement being proved

theorem: a mathematical assertion that can be shown to be true

conjecture: a mathematical assertion proposed to be true, but that has not been proved

proof: a demonstration that a theorem is true

axiom: a statement that is assumed to be true and that can be used as a basis for proving theorems

lemma: a theorem used to prove other theorems

corollary: a proposition that can be proved as a consequence of a theorem that has just been proved

vacuous proof: a proof that $p \rightarrow q$ is true based on the fact that p is false

trivial proof: a proof that $p \rightarrow q$ is true based on the fact that q is true

direct proof: a proof that $p \rightarrow q$ is true that proceeds by showing that q must be true when p is true

proof by contraposition: a proof that $p \rightarrow q$ is true that proceeds by showing that p must be false when q is false

proof by contradiction: a proof that p is true based on the truth of the conditional statement $\neg p \rightarrow q$, where q is a contradiction

exhaustive proof: a proof that establishes a result by checking a list of all possible cases

proof by cases: a proof broken into separate cases, where these cases cover all possibilities

without loss of generality: an assumption in a proof that makes it possible to prove a theorem by reducing the number of cases to consider in the proof

counterexample: an element x such that $P(x)$ is false

constructive existence proof: a proof that an element with a specified property exists that explicitly finds such an element

nonconstructive existence proof: a proof that an element with a specified property exists that does not explicitly find such an element

rational number: a number that can be expressed as the ratio of two integers p and q such that $q \neq 0$

uniqueness proof: a proof that there is exactly one element satisfying a specified property

- Negation $\neg$
- Conjunction $\wedge$
- Disjunction $\vee$
- Implication $\rightarrow$
- Biconditional $\leftrightarrow$

p	$\neg p$
T	F
F	T

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Different Ways of Expressing $p \rightarrow q$

if p , then q

if p , q

p implies q

p only if q

p is sufficient for q

q is necessary for p

a necessary condition for p is q

a sufficient condition for q is p

q when p

q unless $\neg p$

q if p

q whenever p

q follows from p

Converse, Inverse , and Contrapositive

From $p \rightarrow q$ we can form new conditional statements .

- $q \rightarrow p$ is the **converse** of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of
“It raining is a sufficient condition for my not going to town.”

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

TABLE 6 Logical Equivalences.

<i>Equivalence</i>	<i>Name</i>
$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

De Morgan's Laws for Quantifiers

The reasoning in the table shows that:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

The rules for negations for quantifiers are called
De Morgan's laws for quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Quantifications of Two Variables

Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y